

Class 9: Candy Mini Project

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Background

In this mini-project, you will explore FiveThirtyEight's Halloween Candy Dataset.

We will use lots of **ggplot** some basic stats, correlation analysis and PCA to make sense of the landscape of US candy - something hopefully more relocatable than the prote transcriptomics work that we will use these methods throughout the rest of the course.

Data Import

peak with `head()`

```
candy <- read.csv("candy-data.csv", row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crisped	ricewafer
100 Grand	1	0	1	0	0		1
3 Musketeers	1	0	0	0	1		0
One dime	0	0	0	0	0		0
One quarter	0	0	0	0	0		0
Air Heads	0	1	0	0	0		0
Almond Joy	1	0	0	1	0		0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

There are 85 different types of candy in this dataset.

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
candy$fruity
```

```
[1] 0 0 0 0 1 0 0 0 0 1 0 1 1 1 1 1 1 1 0 1 1 0 0 0 0 1 0 0 1 1 1 0 0 1 0 0 0  
[39] 0 0 0 1 0 0 1 1 0 0 0 1 1 0 0 0 0 1 0 0 1 0 1 1 0 1 0 0 1 1 1 1 0 0 1 1 1 0  
[77] 0 0 1 0 1 1 1 0 0
```

```
sum(candy$fruity)
```

```
[1] 38
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is its winpercent value?

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>  
  filter(row.names(candy)=="Hershey's Milk Chocolate") |>  
  select(winpercent)
```

```

                                winpercent
Hershey's Milk Chocolate      56.4905

```

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat", "winpercent"]
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", "winpercent"]
```

```
[1] 49.6535
```

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes, if we did PCA on the data, winpercent would dominate the scale.

```

library("skimr")
skim(candy)

```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q7. What do you think a zero and one represent for the candy\$chocolate column?

```
library("skimr")
skim(candy$chocolate)
```

Table 3: Data summary

Name	candy\$chocolate
Number of rows	85
Number of columns	1
Column type frequency:	
numeric	1
Group variables	None

Variable type: numeric

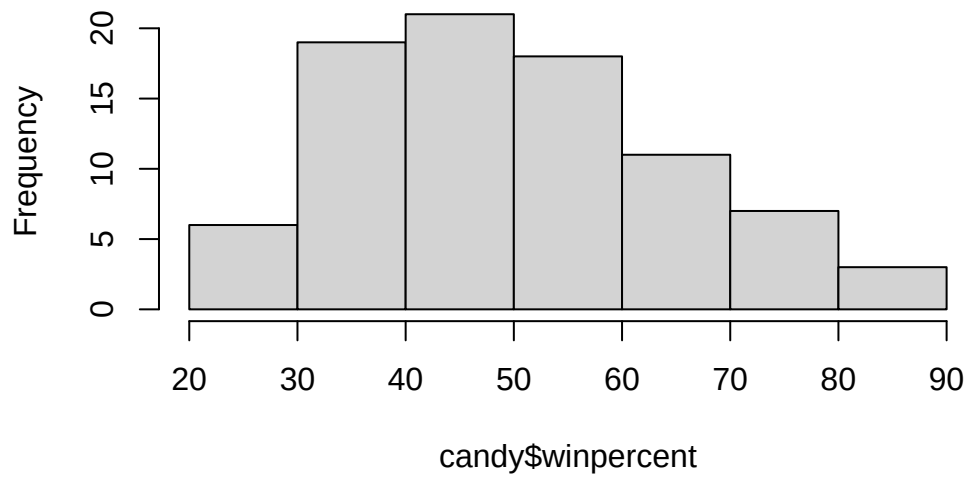
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
data	0	1	0.44	0.5	0	0	0	1	1	

Exploratory Analysis

Q8. Plot a histogram of winpercent values

```
hist(candy$winpercent)
```

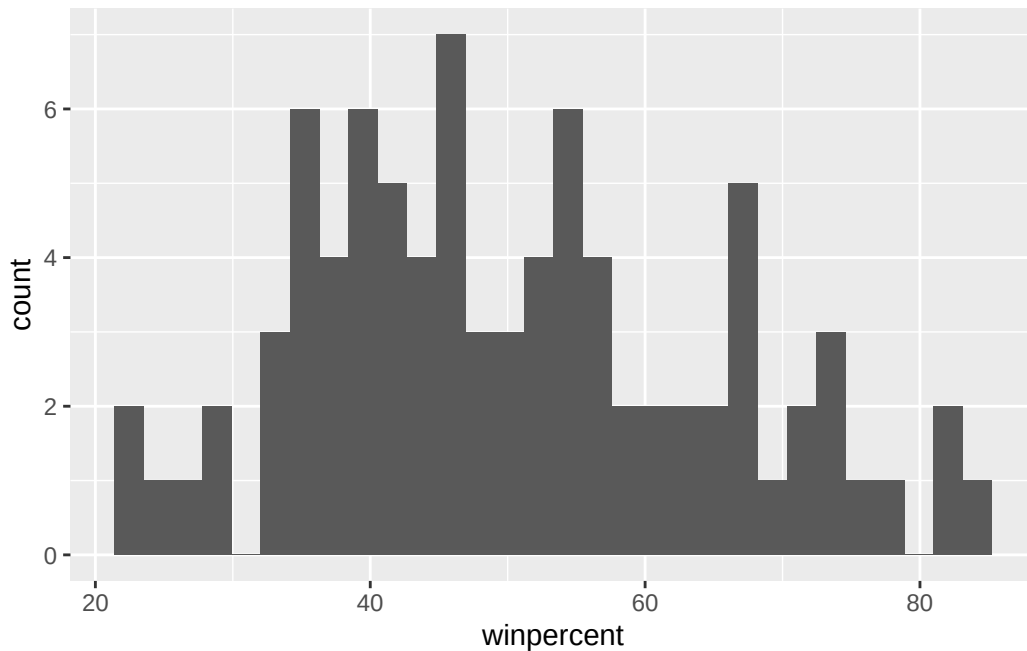
Histogram of candy\$winpercent



```
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_histogram()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Q9. Is the distribution of winpercent values symmetrical?

No the distribution is not symmetrical it is slightly skewed to left.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

Chocolate candy is higher ranked than fruit candy.

```
choc.candy <- candy[candy$chocolate == 1, ]
choc.win <- choc.candy$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```
fruity.win <- candy[candy$fruity == 1,]$winpercent  
mean(fruity.win)
```

```
[1] 44.11974
```

Q12. Is this difference statistically significant?

Chocolate has a smaller p-value than Fruity candy

]

```
chocolate_rank <- candy$winpercent[as.logical(candy$chocolate)]  
candy_rank <- candy$winpercent[as.logical(candy$fruity)]  
  
t.test(x = chocolate_rank, y = candy_rank)
```

Welch Two Sample t-test

```
data: chocolate_rank and candy_rank  
t = 6.2582, df = 68.882, p-value = 2.871e-08  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
 11.44563 22.15795  
sample estimates:  
mean of x mean of y  
 60.92153  44.11974
```

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
y = c("y", "a", "z")  
sort(y)
```

```
[1] "a" "y" "z"
```

```
y
```

```
[1] "y" "a" "z"
```

```
order(y)
```

```
[1] 2 1 3
```

```
ord.ind <- order(candy$winpercent)
head(candy[ord.ind, ])
```

	chocolate	fruity	caramel	peanut	almond	nougat	
Nik L Nip	0	1	0		0	0	
Boston Baked Beans	0	0	0		1	0	
Chiclets	0	1	0		0	0	
Super Bubble	0	1	0		0	0	
Jawbusters	0	1	0		0	0	
Root Beer Barrels	0	0	0		0	0	

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197		0.976
Boston Baked Beans				0	0	0	1	0.313		0.511
Chiclets				0	0	0	1	0.046		0.325
Super Bubble				0	0	0	0	0.162		0.116
Jawbusters				0	1	0	1	0.093		0.511
Root Beer Barrels				0	1	0	1	0.732		0.069

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744
Root Beer Barrels	29.70369

Q14. What are the top 5 all time favorite candy types out of this set?

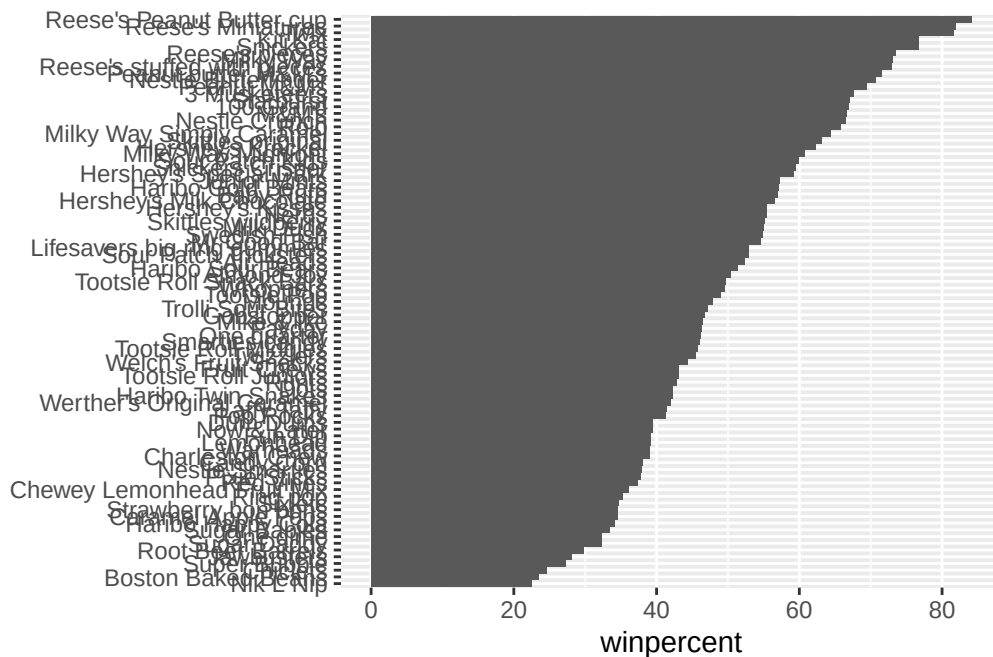
```
head(candy[ord.ind, ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0

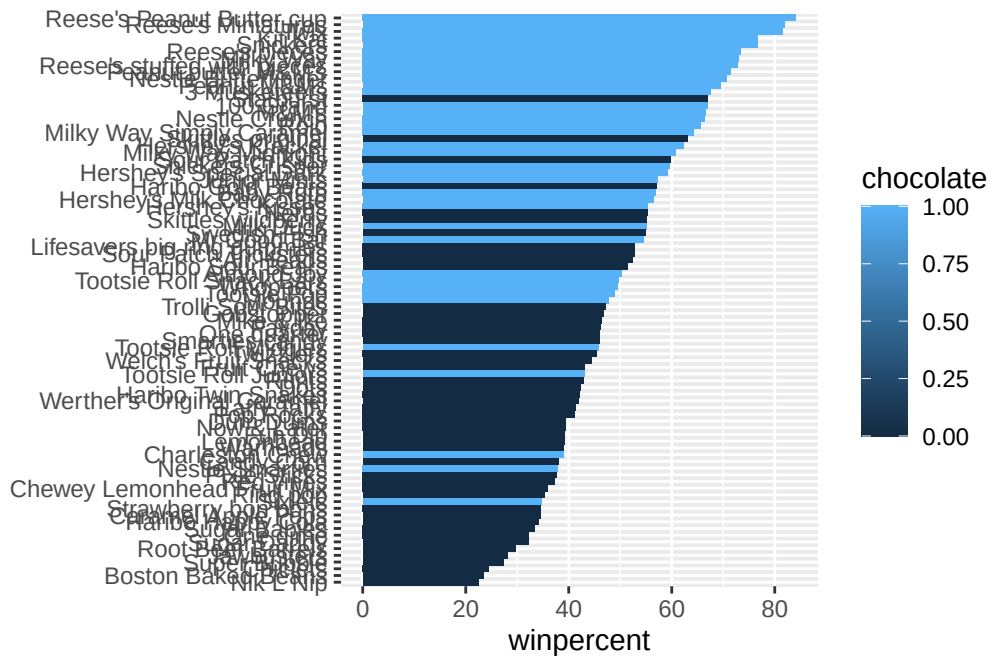
Boston Baked Beans	0	0	0	1	0	
Chiclets	0	1	0	0	0	
Super Bubble	0	1	0	0	0	
Jawbusters	0	1	0	0	0	
	crisped	ricewafer	hard	bar	pluribus	sugarpercent
Nik L Nip		0	0	0	1	0.197
Boston Baked Beans		0	0	0	1	0.313
Chiclets		0	0	0	1	0.046
Super Bubble		0	0	0	0	0.162
Jawbusters		0	1	0	1	0.093
	winpercent					
Nik L Nip	22.44534					
Boston Baked Beans	23.41782					
Chiclets	24.52499					
Super Bubble	27.30386					
Jawbusters	28.12744					

Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy) +
  aes(winpercent,
       reorder(row.names(candy), winpercent)) +
  geom_col() +
  ylab("")
```



```
ggplot(candy) +
  aes(winpercent,
      reorder(row.names(candy), winpercent),
      fill=chocolate) +
  geom_col() +
  ylab("")
```



We need a custom color vector

Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
my_cols <- rep("black", nrow(candy))
#my_cols[1] <- "red" first element is red
my_cols[candy$chocolate==1] <- "chocolate"
my_cols[candy$bar==1] <- "darkgreen"
my_cols[candy$fruity==1] <- "hotpink"
my_cols
```

```
[1] "darkgreen" "darkgreen" "black"      "black"      "hotpink"    "darkgreen"
[7] "darkgreen" "black"      "black"      "hotpink"    "darkgreen" "hotpink"
[13] "hotpink"   "hotpink"   "hotpink"    "hotpink"    "hotpink"    "hotpink"
```

```

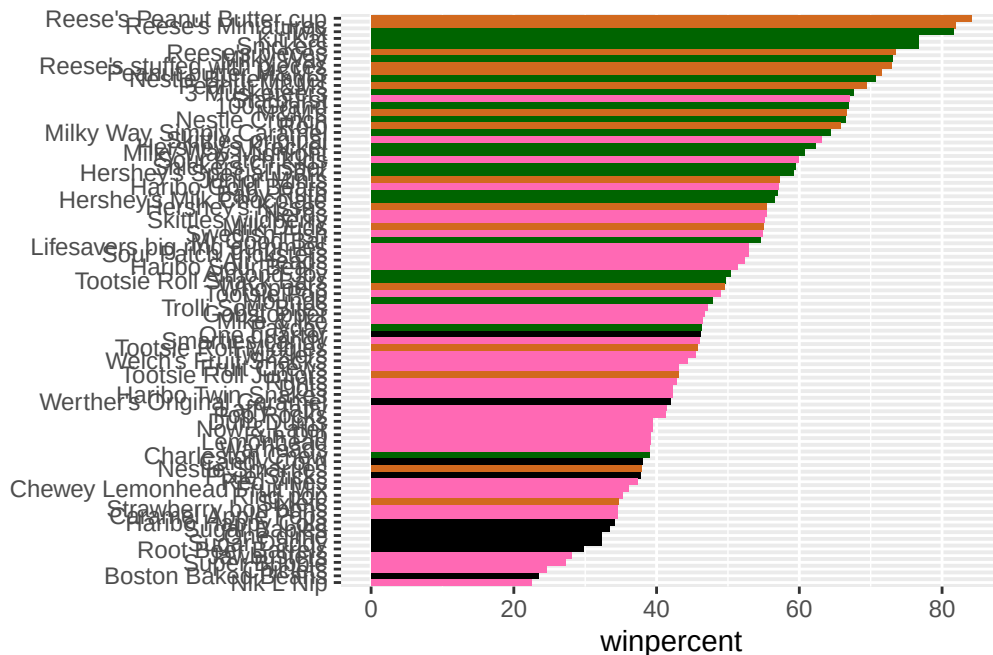
[19] "hotpink"      "black"      "hotpink"    "hotpink"    "chocolate" "darkgreen"
[25] "darkgreen"    "darkgreen"  "hotpink"    "chocolate"  "darkgreen" "hotpink"
[31] "hotpink"      "hotpink"    "chocolate"  "chocolate"  "hotpink"    "chocolate"
[37] "darkgreen"    "darkgreen"  "darkgreen"  "darkgreen"  "darkgreen"  "hotpink"
[43] "darkgreen"    "darkgreen"  "hotpink"    "hotpink"    "darkgreen"  "chocolate"
[49] "black"        "hotpink"    "hotpink"    "chocolate"  "chocolate"  "chocolate"
[55] "chocolate"    "hotpink"    "chocolate"  "black"      "hotpink"    "chocolate"
[61] "hotpink"      "hotpink"    "chocolate"  "hotpink"    "darkgreen"  "darkgreen"
[67] "hotpink"      "hotpink"    "hotpink"    "hotpink"    "black"      "black"
[73] "hotpink"      "hotpink"    "hotpink"    "chocolate"  "chocolate"  "darkgreen"
[79] "hotpink"      "darkgreen"  "hotpink"    "hotpink"    "hotpink"    "black"
[85] "chocolate"

```

```

ggplot(candy) +
  aes(winpercent,
      reorder(row.names(candy), winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")

```



Q17. What is the worst ranked chocolate candy?

Sixlets

Q18. What is the best ranked fruity candy?

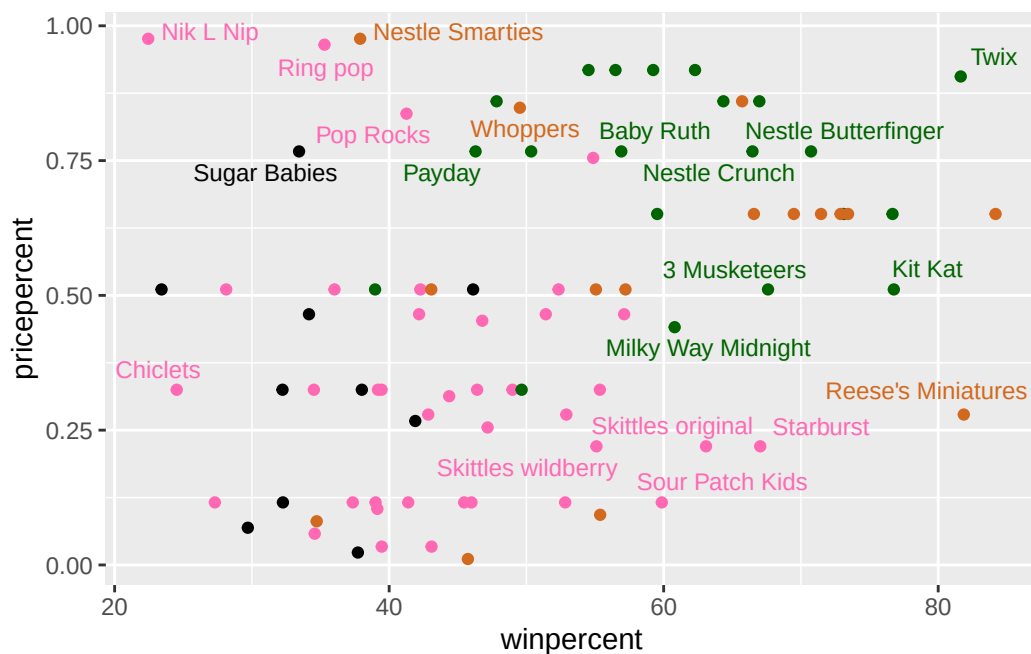
Starburst

Taking a look at pricepercent

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

Warning: ggrepel: 65 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

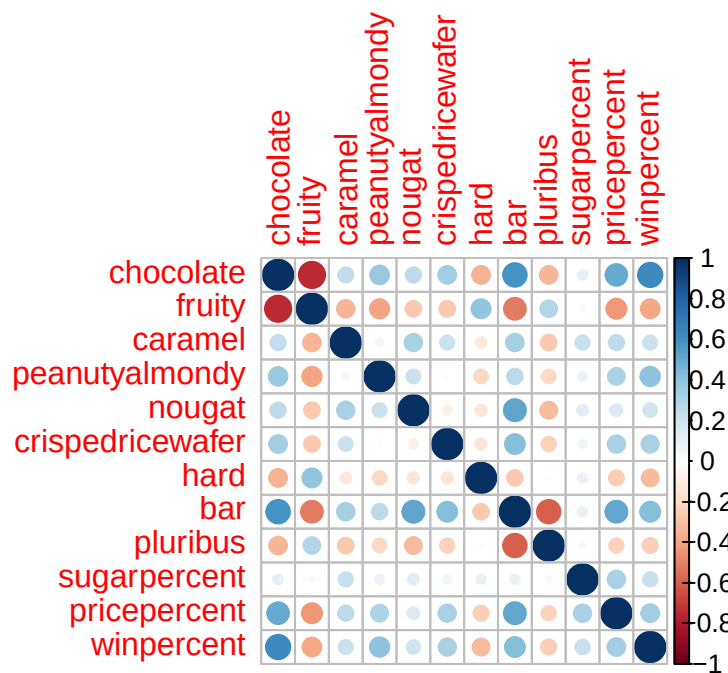
Exploring the correlation structure

```
cij <- cor(candy)
```

```
library(corrplot)
```

corrplot 0.95 loaded

```
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and fruit are anti-correlated.

Q23. Similarly, what two variables are most positively correlated?

Chocolate and bar are the most positively correlated.

Principal Component Analysis

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

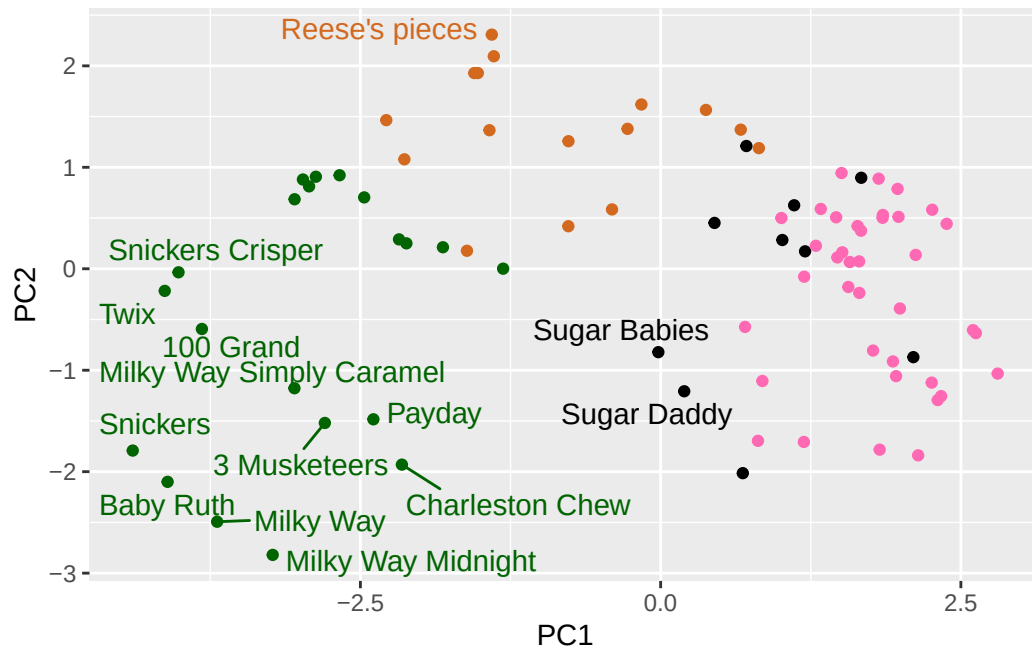
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

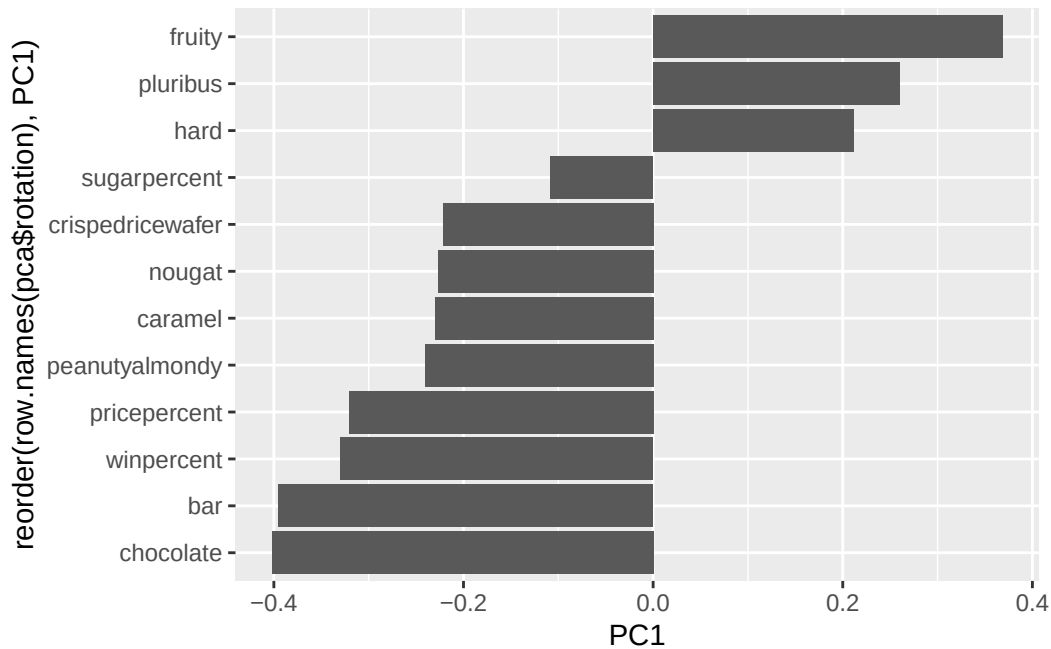
Score Plot...

```
ggplot(pca$x) +
  aes(PC1, PC2, label=row.names(pca$x)) +
  geom_point(col=my_cols)+
  geom_text_repel(max.overlaps = 5, col=my_cols)
```

Warning: ggrepel: 71 unlabeled data points (too many overlaps). Consider increasing max.overlaps



```
ggplot(pca$rotation) +
  aes(PC1,
       reorder(row.names(pca$rotation), PC1)) +
  geom_col()
```



```
#library(plotly)
#ggplotly(p)
```

Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

PC1 is driven positively due to fruity and hard candies while chocolate and bar candies make it negative.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

The combined characteristics that make a winning candy is having the characteristics of chocolate and a bar, they have higher winpercent values and positive correlation as seen on the plots.